

The Chair of Operations Management, in cooperation with Aicher Airport Services, is offering a

Project Study / Interdisciplinary Project

on the topic

Optimizing Airport Transfers for Passengers with Reduced Mobility: Process Design and Evaluation

Scope

Around 1,000 passengers with reduced mobility pass through Munich Airport every day, as inbound, outbound or transit passengers. Aicher Airport Services provides assistance services for these passengers, including accompanying or transporting them from the pick-up point (the meeting point for outbound passengers, the aircraft or gate for inbound and transit passengers) to the drop-off point (the taxi rank or public transport stop for inbound passengers, the aircraft or gate for outbound and transit passengers). Various vehicles are used for transportation, including vans on the apron, wheelchairs, and electric gate taxis in the terminal. Currently, during end-to-end assistance, a single service person accompanies the passenger from the pick-up point to the drop-off point. This process will be compared to alternative ones in the project study.

One such alternative is a zonal approach, in which, instead of a single service person, different specialised service workers accompany passengers along their journey. For example, the first part of the journey would involve taking a van from the plane to the terminal, while the second part would involve transferring within the terminal.

The project study will analyse and describe the current process and develop alternative process models. Furthermore, based on data provided by Aicher Airport Services, which may be supplemented by additional data, students must evaluate the key performance indicators and costs of the various processes. This may involve simulation and optimisation approaches. The aim is to analyse the existing process models and develop new process models that reduce costs per passenger served and other key performance indicators.

Requirements

- Interest in air traffic and airports
- Knowledge of the fundamentals of Operations Research
- Basic knowledge in Simulation (e.g., through the module MOS) and/or database systems
- Good IT and programming skills, preferably Python
- Reliability and resilience

Start and duration: As soon as possible; duration: 3 months full-time, longer for part-time work

Language: German or English

Number of participants for each project study: 2-4

Supervisor: Deepthi Cherullil Padinjaroot (MSc.)

Contact: If you are interested, please contact deepthi.cherullil@tum.de

Supervisor Aicher Ambulanz: Christian Herrmann (Abteilungsleiter Disposition und Prozesse)